Tanmay Bhuskute

+1 (732) 858-3027 | tanmay.v.bhuskute@gmail.com | github.com/Tanmayb05 | linkedin.com/in/tanmay-bhuskute
Cincinnati, OH

SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) building LLM-powered and agentic systems: retrieval-augmented pipelines, tool-calling agents, and evaluation harnesses in Python, LangChain, and Gemini/Vertex AI, backed by production cloud engineering experience.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, TypeScript, SQL, C/C++, Go.

AI & Agentic Systems: LangChain, RAG, LLM APIs (Gemini, Vertex AI, GPT-4o), Agentic Tool-Calling, MCP, Evals, PyTorch, TensorFlow, scikit-learn.

Data & Retrieval: Vector Databases, pandas, NumPy, PostgreSQL, REST APIs.

Web & Backend: React, Node.js, Django, Flask, FastAPI.

Developer Tools: Git, GitLab CI/CD, AWS, Docker, Kubernetes, ArgoCD, Helm, Terraform, Linux, Bash.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Built provider-agnostic LLM abstraction layer (Gemini, GPT-4o, Llama, Qwen) and wrote structured evals demonstrating **25.7x latency reduction** while preserving task-completion fidelity.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing live infrastructure issues across Linux, EC2, databases, and Teamcenter X services.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%**.
- Migrated **40+ production environments** from process-based deployments to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

PROJECTS AND RESEARCH

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG pipeline** over Google Gemini 2.5-flash, forcing citation-backed answers to eliminate hallucination in high-stakes healthcare eligibility decisions.
- Designed **17-tool MCP server** exposing structured LLM tool-calling for PA extraction, patient-readiness evaluation, and cross-payer comparison, consumable by external AI agents.

Spendora – Conversational AI for Expense Insights [LangChain, HuggingFace, Mistral, Gemma] Jun 2025

- Built LLM-powered finance assistant with LangChain, implementing **RAG** and custom toolchains that automated manual expense categorization and surfaced spending patterns from raw transaction data via conversational agents with memory and tool use.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).